Multi-Stage Optimization for Supporting Lines in Logistics Companies

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Outline

- Motivation
- The proposed approach
- Experiment
- Discussion
- Conclusion & Future

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Motivation

- Logistics distribution demand is continuously increasing
  - Growth of logistics and e-commerce increases delivery complexity
  - Efficient route planning is critical for reducing operational costs

- Real-world logistics operations differ from Vehicle Routing Problem (VRP) models
  - Fixed-route structures and service restrictions are rarely considered in VRP studies

- Route planning is performed manually
  - Route planning is a time-consuming
  - Route planning is highly dependent on human experience

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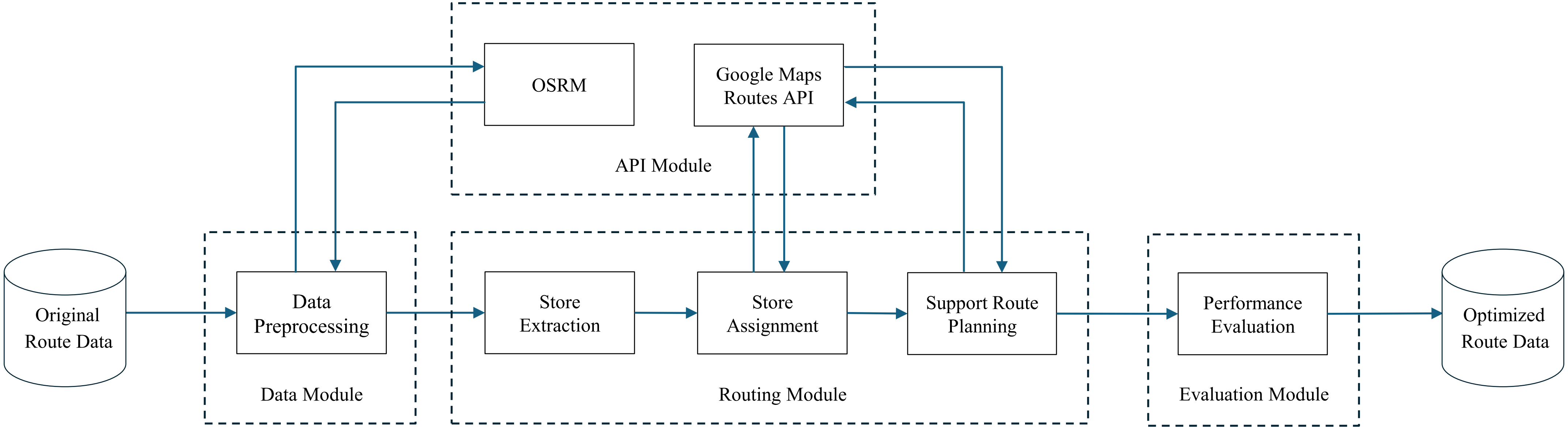
The proposed approach

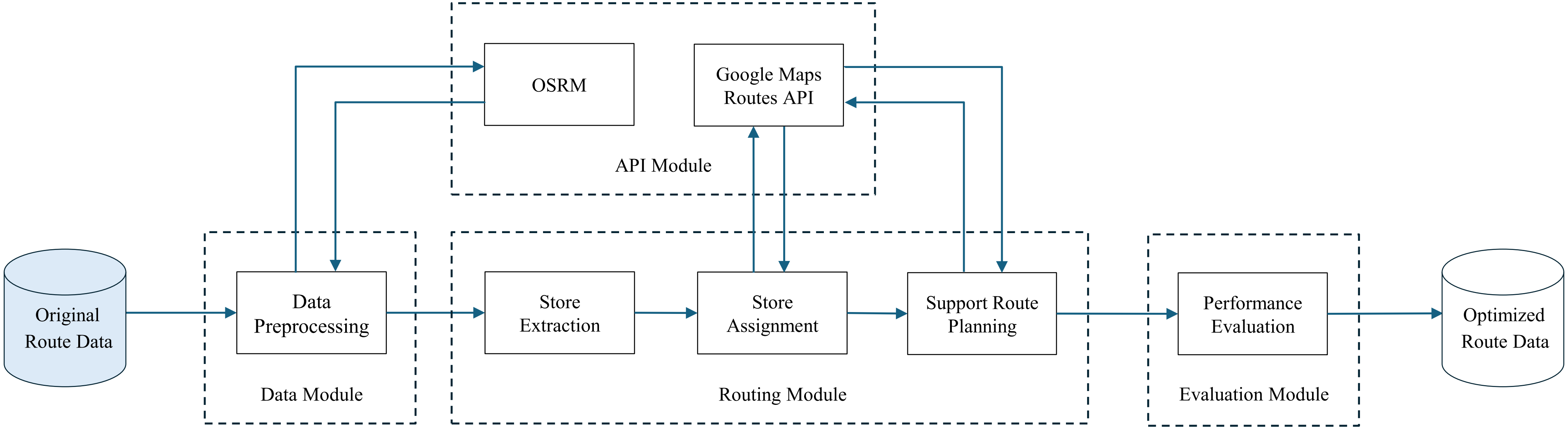
Multi-stage Operationally Constrained VRP Framework composed of four main modules

1. API Module
   - Provide distance and travel-time data through Open Source Routing Machine (OSRM) and Routes API
2. Data Module
   - Preprocess raw routing data for route planning
3. Routing Module
   - Perform store extraction, assignment, and support route planning under operational constraints
4. Evaluation Module
   - Provide route visualization and performance evaluation using multiple routing metrics

The proposed approach



The proposed approach



Original Route Data

The experimental dataset consists of 26 working days of original route data

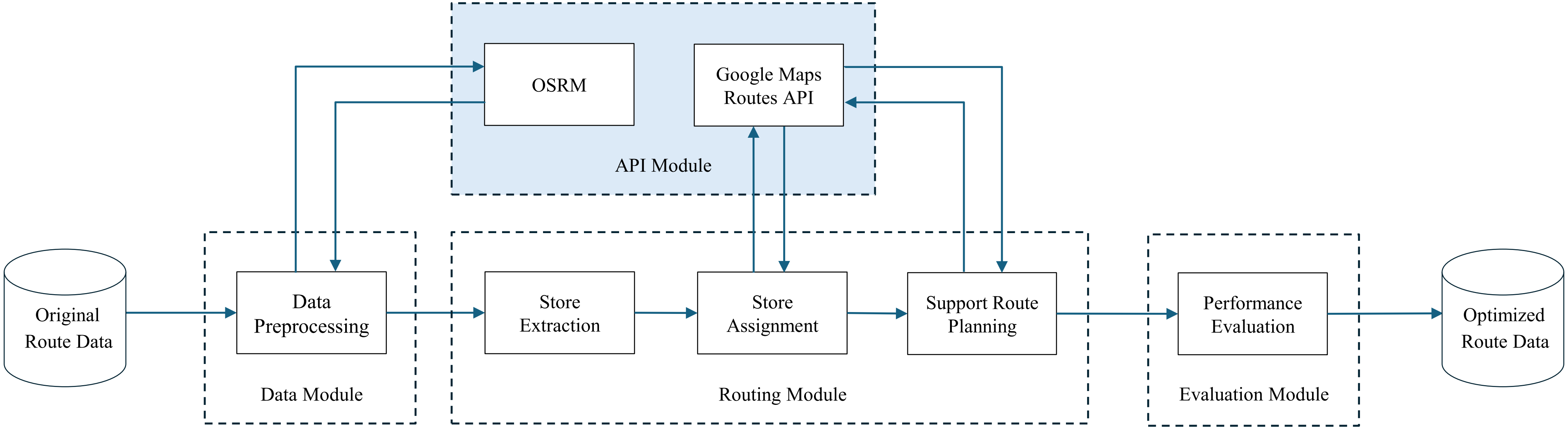
不可大車	車次	店名	表定時間	預定時間	貨量	裝載率
	2M	林口D C			3.16	0.438888889
False	2M01	機場捷運店	2022-12-06 18:20:00	2022-12-06 18:20:00	0.85	
False	2M02	高鐵二店	2022-12-06 18:30:00	2022-12-06 18:30:00	1.52	
False	2M03	台鐵一店	2022-12-06 19:00:00	2022-12-06 19:00:00	0.79	
False	2MDC	林口D C	2022-12-02 19:50:00	2022-12-02 19:50:00	0	
	2N	林口D C			11.56	0.802777778
False	2N01	林口麗林店	2022-12-07 06:00:00	2022-12-07 06:00:00	1.06	
False	2N02	林口新都店	2022-12-07 06:30:00	2022-12-07 06:30:00	0.85	
False	2N03	林口影城店	2022-12-07 06:45:00	2022-12-07 06:45:00	0.14	
False	2N04	林口菁英店	2022-12-07 07:00:00	2022-12-07 07:00:00	1.86	
False	2N05	龜山大義店	2022-12-07 07:15:00	2022-12-07 07:15:00	0.55	
False	2N06	龜山甲子店	2022-12-07 07:40:00	2022-12-07 07:40:00	0.49	
False	2N07	龜山善捷店	2022-12-07 07:50:00	2022-12-07 07:50:00	0.69	
False	2N08	龜山新都店	2022-12-07 08:20:00	2022-12-07 08:20:00	1.35	
False	2N09	龜山新城店	2022-12-07 08:35:00	2022-12-07 08:35:00	0.19	
False	2N10	龜山庚勤店	2022-12-07 08:55:00	2022-12-07 08:55:00	1.41	
False	2N11	龜山長明店	2022-12-07 09:15:00	2022-12-07 09:15:00	1	
False	2N12	龜山庚德店	2022-12-07 09:30:00	2022-12-07 09:30:00	1.25	
False	2N13	龜山庚行店	2022-12-07 09:45:00	2022-12-07 09:45:00	0.49	
False	2N14	龜山大湖店	2022-12-07 10:20:00	2022-12-07 10:20:00	0.23	
False	2NDC	林口D C	2022-12-07 10:31:00	2022-12-07 10:31:00	0	

Original Route Data

Daily demand fluctuations may lead to route overloading

不可大車	車次	店名	表定時間	預定時間	貨量	裝載率
	3D	林口ＤＣ			8.45	1.173611111
False	3D01	八里富埔店	2022-12-06 16:50:00	2022-12-06 16:50:00	0.62	
False	3D02	中永店	2022-12-06 18:00:00	2022-12-06 18:00:00	0.62	
False	3D03	永興店	2022-12-06 18:15:00	2022-12-06 18:15:00	0.52	
False	3D04	新奇岩店	2022-12-06 18:30:00	2022-12-06 18:30:00	0.77	
False	3D05	三合店	2022-12-06 18:45:00	2022-12-06 18:45:00	0.84	
False	3D06	榮岩店	2022-12-06 19:00:00	2022-12-06 19:00:00	0.73	
False	3D07	中央南路店	2022-12-06 19:15:00	2022-12-06 19:15:00	0.88	
False	3D08	東陽店	2022-12-06 19:30:00	2022-12-06 19:30:00	1.18	
False	3D09	立賢店	2022-12-06 19:45:00	2022-12-06 19:45:00	1.06	
False	3D10	社子島店	2022-12-06 20:00:00	2022-12-06 20:00:00	1.23	
False	3DDC	林口ＤＣ	2022-12-06 20:22:00	2022-12-06 20:22:00	0	
	3E	林口ＤＣ			5.31	0.7375
False	3E01	林口醒吾店	2022-12-06 17:30:00	2022-12-06 17:30:00	0.33	
False	3E02	五股登科店	2022-12-06 17:50:00	2022-12-06 17:50:00	0.96	
False	3E03	德惠店	2022-12-06 18:30:00	2022-12-06 18:30:00	0.79	
False	3E04	權勝店	2022-12-06 18:50:00	2022-12-06 18:50:00	0.36	
False	3E05	錦民店	2022-12-06 19:10:00	2022-12-06 19:10:00	0.95	
False	3E06	民佳店	2022-12-06 19:30:00	2022-12-06 19:30:00	0.33	
False	3E07	松好店	2022-12-06 19:50:00	2022-12-06 19:50:00	0.3	
False	3E08	松茂店	2022-12-06 20:10:00	2022-12-06 20:10:00	1.29	
False	3EDC	林口ＤＣ	2022-12-06 20:31:00	2022-12-06 20:31:00	0	

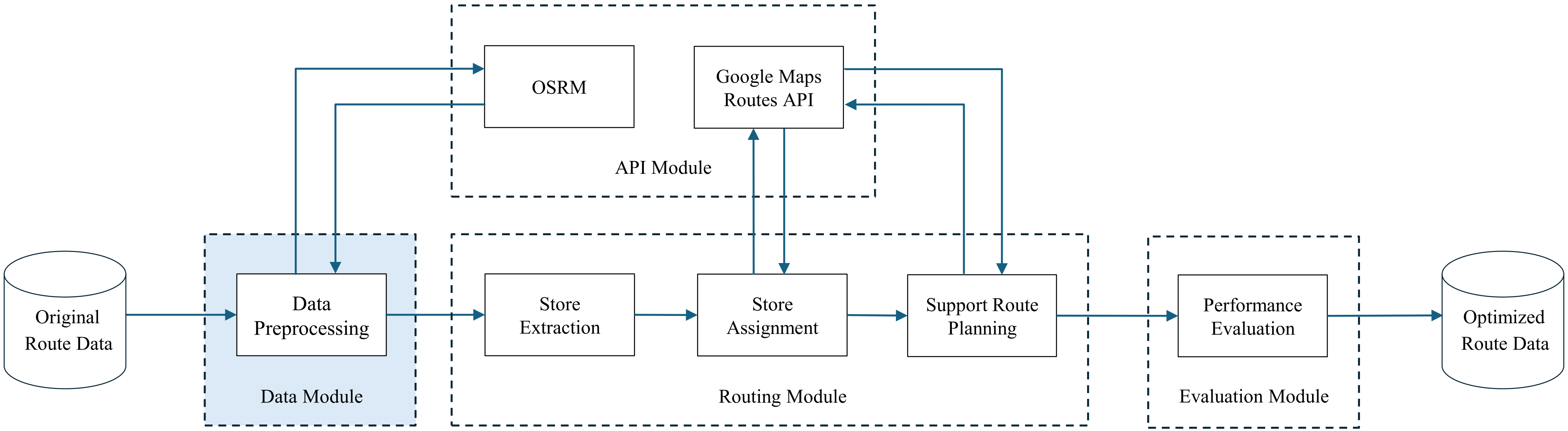
The proposed approach



API Module

1. Open Source Routing Machine (OSRM)
   - Provides distance and travel time matrices for efficient route planning and optimization
2. Google Maps Routes API
   - Validate and evaluate routing results using real-world road network data

The proposed approach

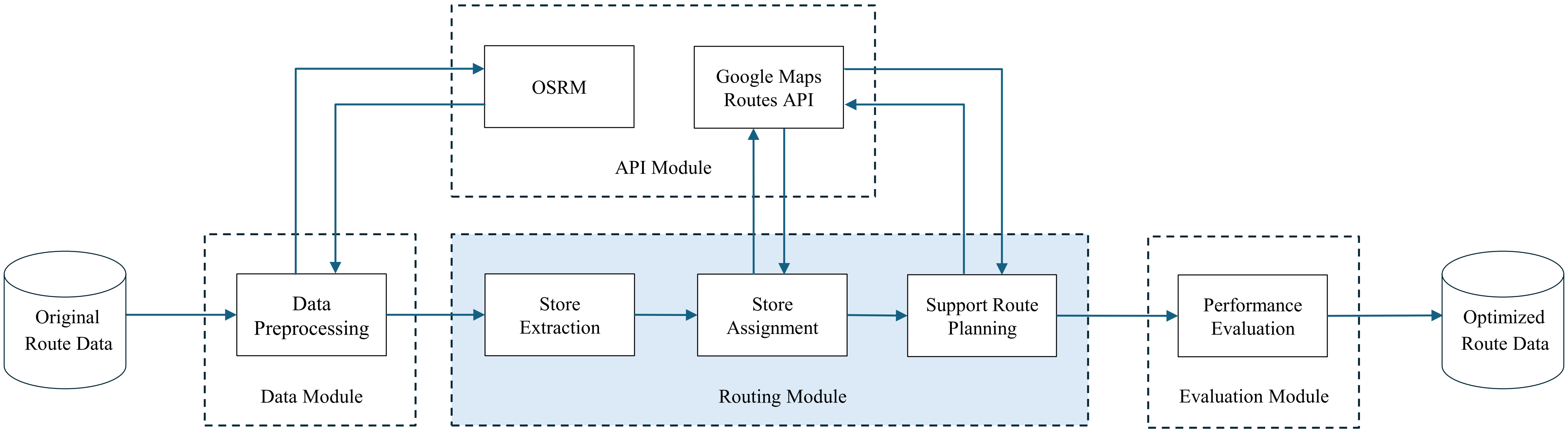


Data Preprocessing

Data preprocessing consists of data cleaning and data transformation processes

1. Data Cleaning
   - ➢ Handle missing or incomplete store attributes
2. Data Transformation
   - ➢ Convert the data into a structured format for routing optimization

The proposed approach



Routing Module

The proposed framework is developed by considering the constraints of the logistics company, and mainly consists of three stages

1. Store Extraction
2. Store Assignment
3. Support Route Planning

- The routing process in the proposed framework is formulated as a Vehicle Routing Problem (VRP)

VRP

Vehicle Routing Problem (VRP)[1]

- NP-hard problem in combinatorial optimization and integer programming
- The objective is to ensure that all customer demands are satisfied while complying with operational constraints and minimizing the overall cost
- Constraints
  - Each vehicle starts and ends its route at the depot
  - Each customer is visited exactly once and served by exactly one vehicle
  - The demand of every customer must be satisfied

[1] P. Toth and D. Vigo, *The vehicle routing problem*. SIAM, 2002.

VRP

- Variants
  - Capacitated Vehicle Routing Problem (CVRP)
  - Vehicle Routing Problem with Time Windows (VRPTW)
  - Vehicle Routing Problem with Private Fleet and Common Carrier (VRPPC)
  - etc.

Constraints

➢The proposed optimization model is subject to the following constraints

- Capacity Constraint

$$\sum_{i \in r_k} v_i \leq Q_k,$$

- Time Window Constraint

$$e_j \leq t_j \leq l_j$$

- Duration Constraint

$$T_k \leq T_{max}$$

Constraints

The constraints considered in this study are defined as follows

1. The main routes are defined with a fixed sequence of stores, and the ordering cannot be modified
2. The extent of route modification is minimized while preserving the main route structure
3. The vehicle capacity must not be exceeded
4. The time window is defined as 1 hour before and 30 minutes after the scheduled time
5. The total route duration must not exceed 5 hours

Objective Function

- The objective function is formulated as a hierarchical objective
  - Minimize Number of Support Routes

$$f_{vehicle}(S) = \sum_{h \in H} \sum_{\substack{j \in N \\ (j \neq 0)}} x_{0,j}^{h}$$

  - Minimize Total Distance

$$f_{travel}(S) = \sum_{k \in K} \sum_{i \in N} \sum_{\substack{j \in N \\ (j \neq i)}} d_{i,j} \cdot x_{i,j}^{k}$$

Store Extraction

- Address vehicle overloading caused by demand fluctuations by removing selected stores from overloaded main routes to satisfy capacity and time window constraints

- Cooperative Co-evolutionary Genetic Algorithm (CCGA)[2]
  - Decompose the problem into multiple smaller subproblems
  - Evolve each subproblem independently using Genetic Algorithms
  - Combine partial solutions and evaluate the complete solution
  - $fitness = w_1 \cdot f_{vehicle}(S) + w_2 \cdot |R| + f_{travel}(S)$

[2] X. Ma *et al.*, "A survey on cooperative co-evolutionary algorithms," *IEEE Transactions on Evolutionary Computation,* vol. 23, no. 3, pp. 421–441, 2018.

Store Assignment

- Insert candidate stores into main routes with available capacity to improve utilization
- Hybrid Ant Colony Optimization (HACO)[3]
  - Combine Ant Colony Optimization (ACO) and Variable Neighborhood Descent (VND)
  - ACO constructs solutions by simulating pheromone-based communication among ants
  - VND improves solutions by sequentially exploring multiple neighborhood to escape local optima

[3] Q. Ding, X. Hu, L. Sun, and Y. Wang, "An improved ant colony optimization and its application to vehicle routing problem with time windows," *Neurocomputing,* vol. 98, pp. 101–107, 2012.

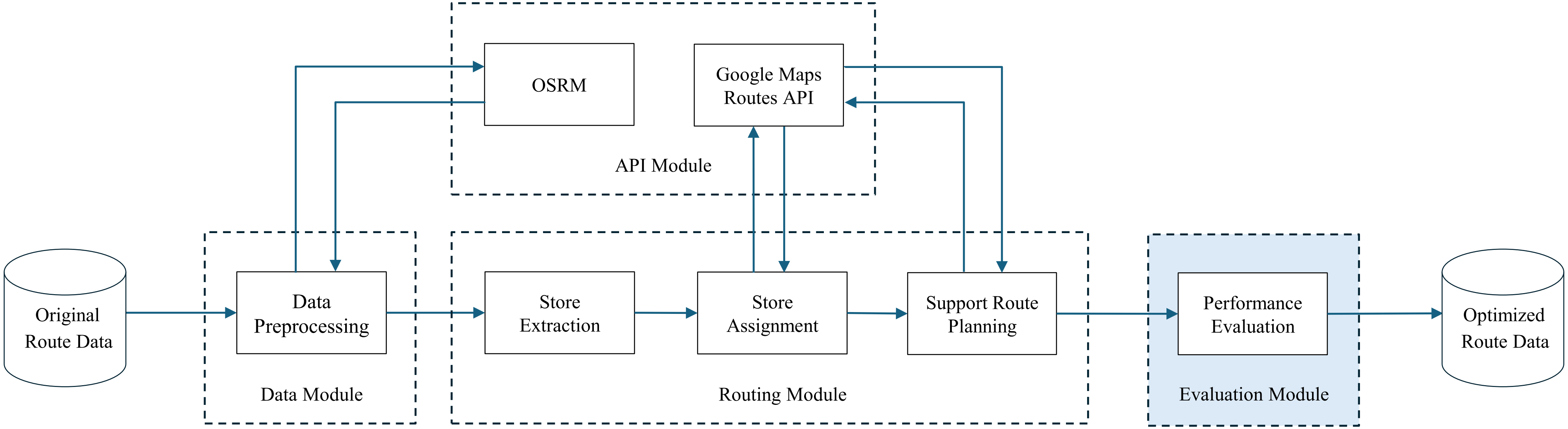
Support Route Planning

- Perform routing for stores that cannot be assigned to main routes

- Multiple Ant Colony System (MACS)[4]
  - ACS-VEI minimizes the number of vehicles by improving store coverage during route construction
  - ACS-DIST minimizes total travel distance by applying VND for local optimization after solution construction.

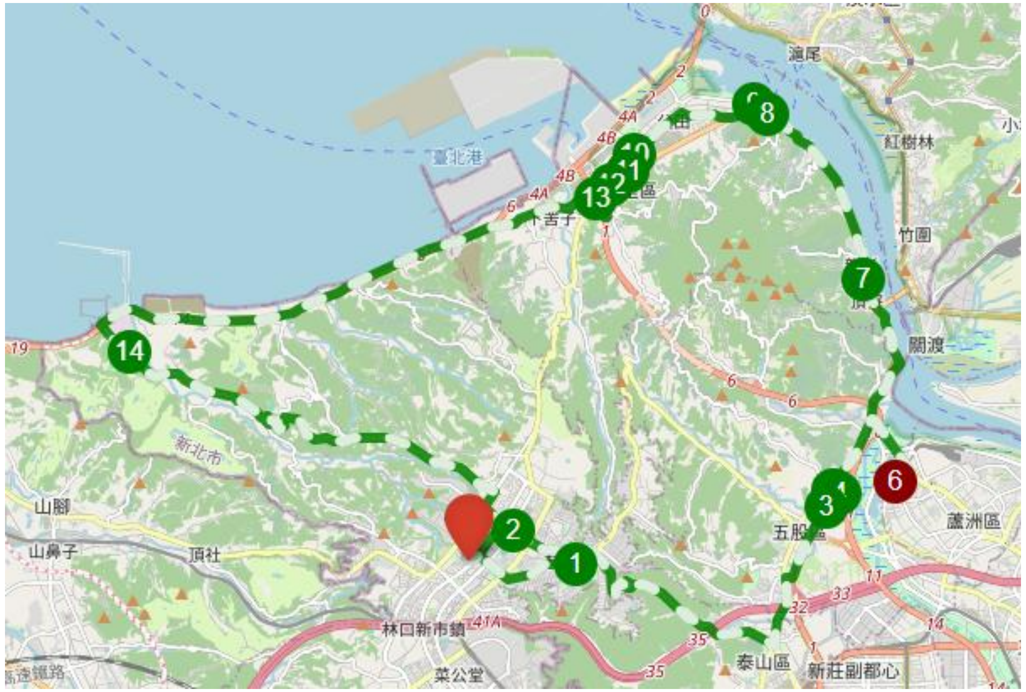
[4] L. M. Gambardella, É. Taillard, and G. Agazzi, "MACS-VRPTW: A multiple ant colony system for vehicle routing problems with time windows," ed: Istituto Dalle Molle Di Studi Sull Intelligenza Artificiale, 1999

The proposed approach



Evaluation Module

- The routing results are visualized on map-based visualization platforms

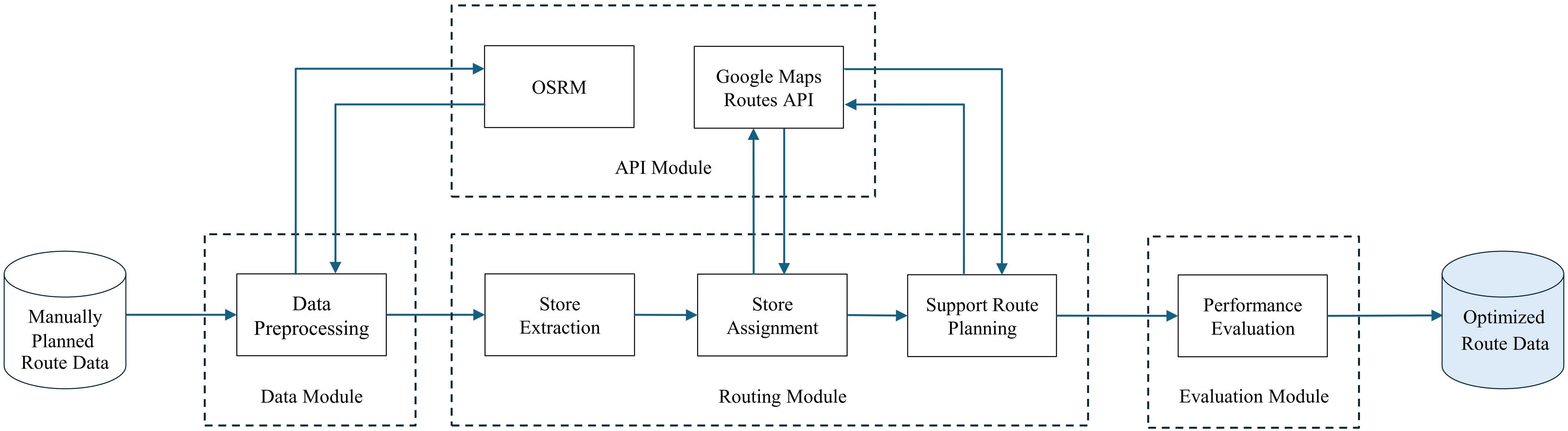


Evaluate Module

The following metrics are used to evaluate the performance of the proposed VRP framework

1. Number of Vehicles
2. Number of Support Vehicles
3. Total Distance
4. Average Load Rate
5. On Time Rate
6. Runtime

The proposed approach



Optimized Route Data

車次	ID	店名	表定時間	最早抵達時間	預定時間	最晚抵達時間	貨量	裝載率	時間(hr)	距離(km)
2M		林口DC					3.16	0.4388889	1.7036	44.05
2M01	1001021597	機場捷運店	2022-12-06T18:20:00	2022-12-06T17:20:00	2022-12-06T18:20:00	2022-12-06T18:50:00	0.85			
2M02	1001018520	高鐵二店	2022-12-06T18:30:00	2022-12-06T17:30:00	2022-12-06T18:36:57	2022-12-06T19:00:00	1.52			
2M03	1001018687	台鐵一店	2022-12-06T19:00:00	2022-12-06T18:00:00	2022-12-06T18:52:36	2022-12-06T19:30:00	0.79			
2MDC		林口DC					0			
2N		林口DC					11.56	0.8027778	5.0781	33.39
2N01	1001015407	林口麗林店	2022-12-07T06:00:00	2022-12-07T05:00:00	2022-12-07T06:00:00	2022-12-07T06:30:00	1.06			
2N02	1001022531	林口新都店	2022-12-07T06:30:00	2022-12-07T05:30:00	2022-12-07T06:21:27	2022-12-07T07:00:00	0.85			
2N03	1001022230	林口影城店	2022-12-07T06:45:00	2022-12-07T05:45:00	2022-12-07T06:35:11	2022-12-07T07:15:00	0.14			
2N04	1001017041	林口菁英店	2022-12-07T07:00:00	2022-12-07T06:00:00	2022-12-07T06:51:43	2022-12-07T07:30:00	1.86			
2N05	1001019267	龜山大義店	2022-12-07T07:15:00	2022-12-07T06:15:00	2022-12-07T07:10:18	2022-12-07T07:45:00	0.55			
2N06	1001021576	龜山甲子店	2022-12-07T07:40:00	2022-12-07T06:40:00	2022-12-07T07:35:53	2022-12-07T08:10:00	0.49			
2N07	1001021558	龜山善捷店	2022-12-07T07:50:00	2022-12-07T06:50:00	2022-12-07T07:53:11	2022-12-07T08:20:00	0.69			
2N08	1001017360	龜山新都店	2022-12-07T08:20:00	2022-12-07T07:20:00	2022-12-07T08:12:37	2022-12-07T08:50:00	1.35			
2N09	1001016068	龜山新城店	2022-12-07T08:35:00	2022-12-07T07:35:00	2022-12-07T08:22:30	2022-12-07T09:05:00	0.19			
2N10	1001022088	龜山庚勤店	2022-12-07T08:55:00	2022-12-07T07:55:00	2022-12-07T08:51:35	2022-12-07T09:25:00	1.41			
2N11	1001021758	龜山長明店	2022-12-07T09:15:00	2022-12-07T08:15:00	2022-12-07T09:15:56	2022-12-07T09:45:00	1			
2N12	1001018725	龜山庚德店	2022-12-07T09:30:00	2022-12-07T08:30:00	2022-12-07T09:45:07	2022-12-07T10:00:00	1.25			
2N13	1001020093	龜山庚行店	2022-12-07T09:45:00	2022-12-07T08:45:00	2022-12-07T10:03:10	2022-12-07T10:15:00	0.49			
2N14	1001016413	龜山大湖店	2022-12-07T10:20:00	2022-12-07T09:20:00	2022-12-07T10:32:57	2022-12-07T10:50:00	0.23			
2NDC		林口DC					0			

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Experiment

Item	Details
CPU	Intel® Core™ i7-13700 5.20 GHz
GPU	Intel® UHD graphics 770
RAM	24GB
SSD	1TB HDD
OS	Windows 11 專業版
Python	3.12.6

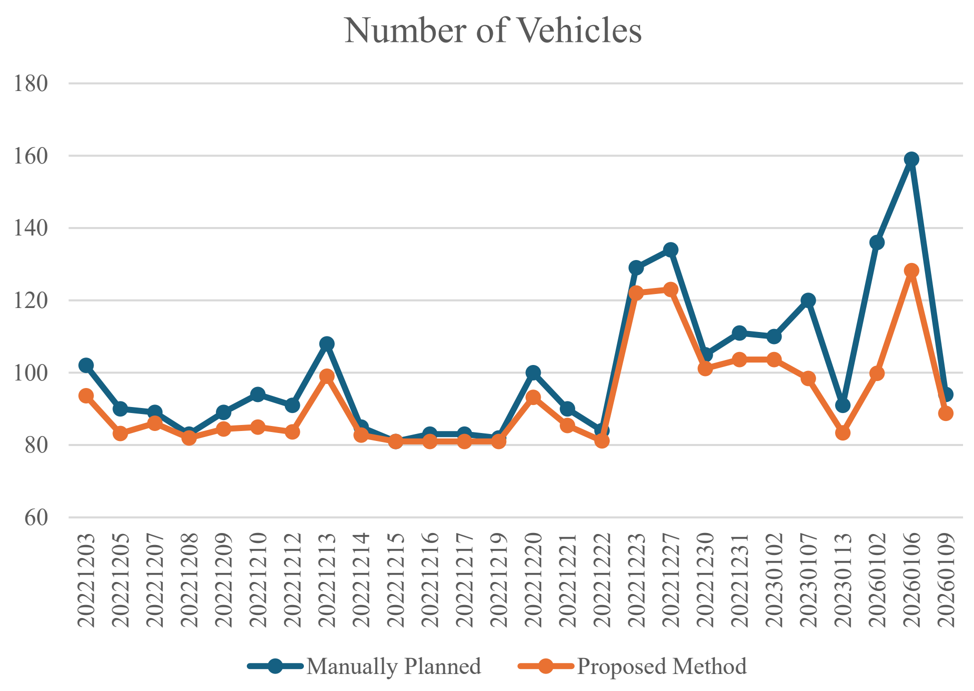
Experiment Setting

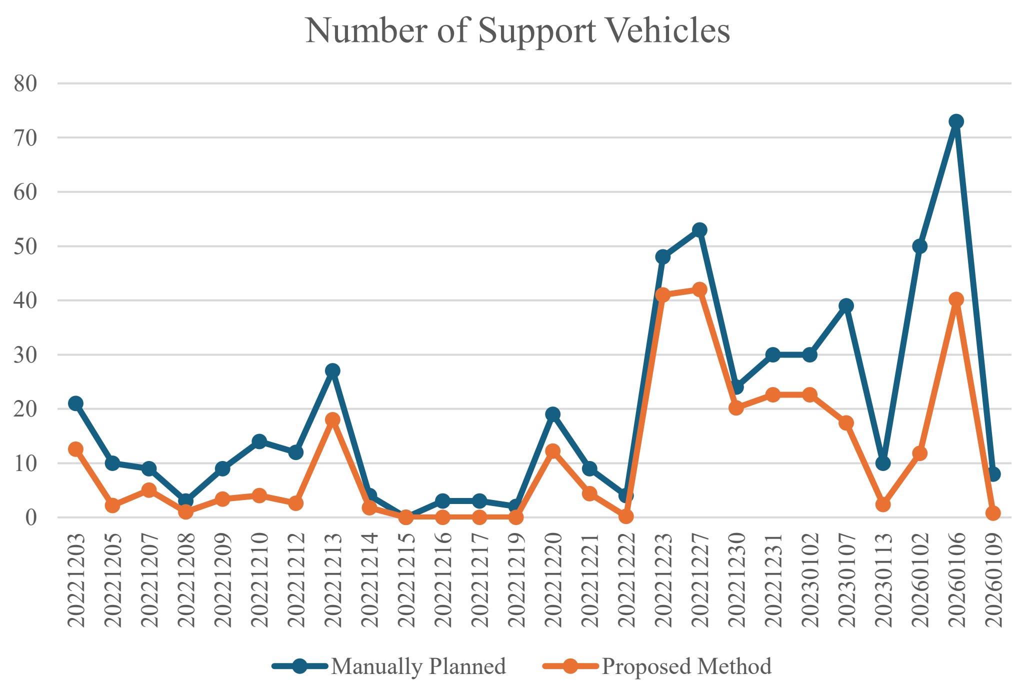
CCGA		HACO		MACS	
Generation	200	Iteration	200	Time Limit (s)	100
Subpopulation	20	Number of Ants	50	Number of Ants	50
Cross Rate	0.8	Pheromone (α)	1	Pheromone (α)	1
Mutate Rate	0.1	Heuristic (β)	1	Heuristic (β)	7
Elite Rate	0.1	Evaporation (ρ)	0.7	Evaporation (ρ)	0.5

Experiment Result

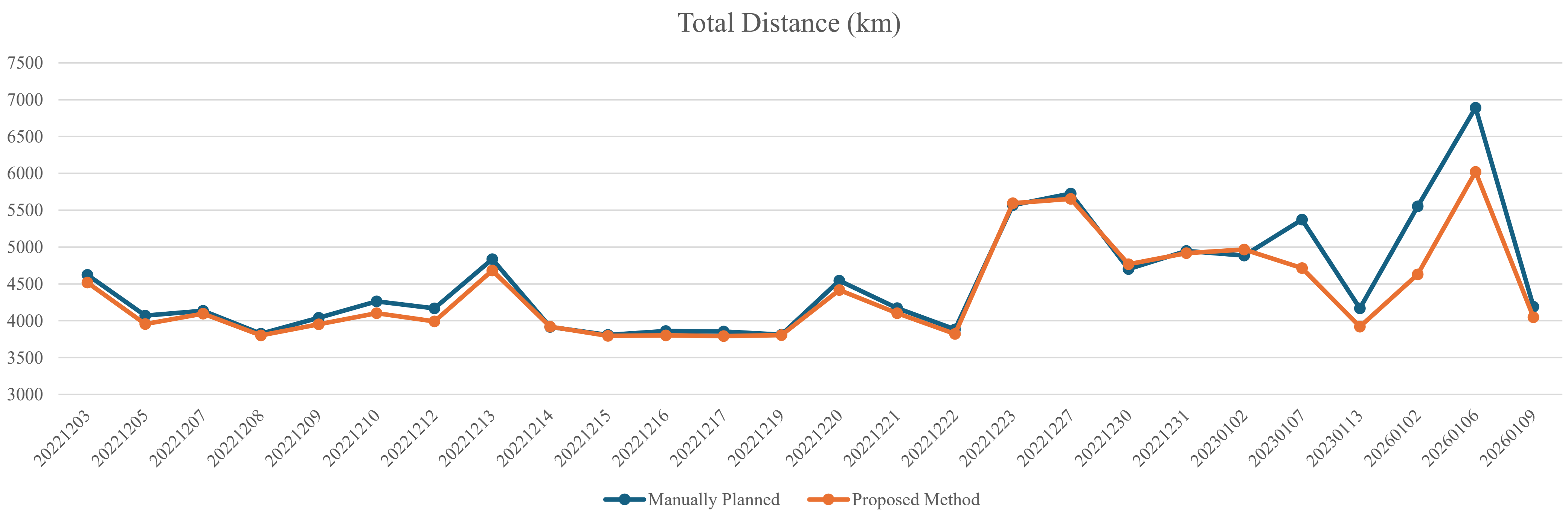
	Proposed Method	Manually Planned
Number of Vehicles	92.94	100.88
Number of Support Vehicles	11.14	19.77
Total Distance (km)	4375.10	4531.51
Average Load Rate	0.89	0.83
On Time Rate	0.99	0.94
Runtime (min)	14	90

Experiment Result

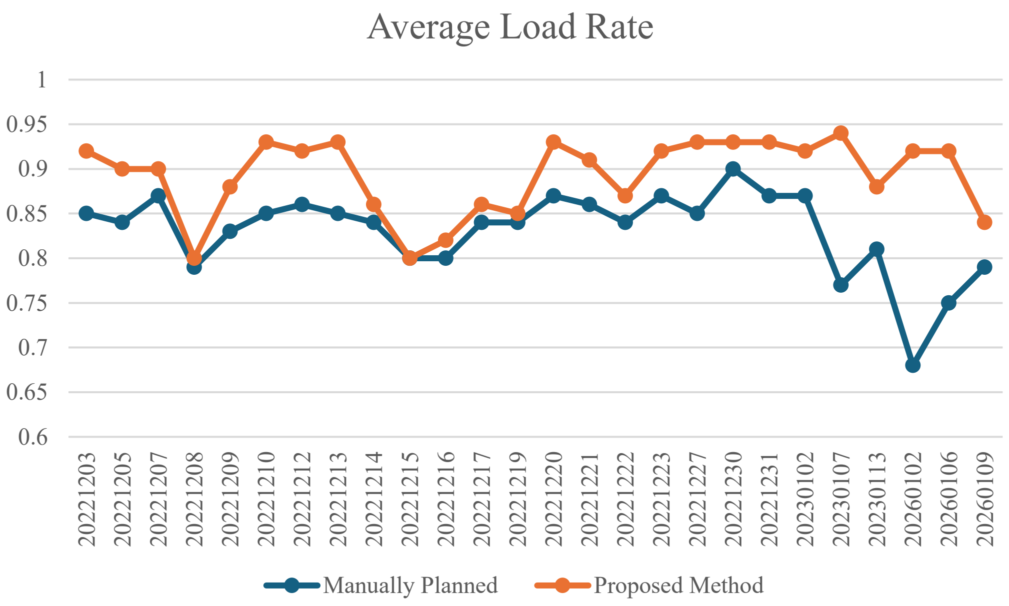


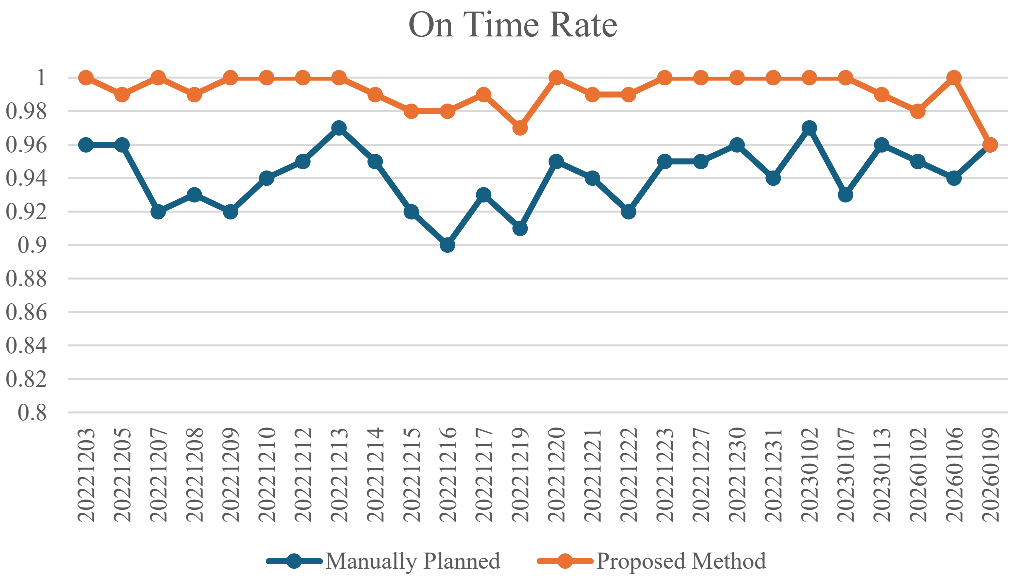


Experiment Result



Experiment Result





Ablation Study

	Full Method	Random Extraction	HACO Without VND	MACS Without VND
Number of Vehicles	92.94	94.24	93.02	93.25
Number of Support Vehicles	11.14	12.43	11.22	11.44
Total Distance (km)	4375.10	4524.44	4382.90	4,400.13
Average Load Rate	0.89	0.88	0.89	0.89
On Time Rate	0.99	0.99	0.99	0.99
Runtime (min)	14	8	13	13

Ablation Study

- Wilcoxon Signed-Rank Test
  - Significant at α=0.05

	Metric	**p-value**
Random Extraction	Number of Vehicles	0.00001
	Number of Support Vehicles	0.00001
	Total Distance	0.00004
HACO Without VND	Number of Vehicles	0.46608
	Number of Support Vehicles	0.48388
	Total Distance	0.01513
MACS Without VND	Number of Vehicles	0.00572
	Number of Support Vehicles	0.00572
	Total Distance	0.00129

Baseline - HGA-SIH

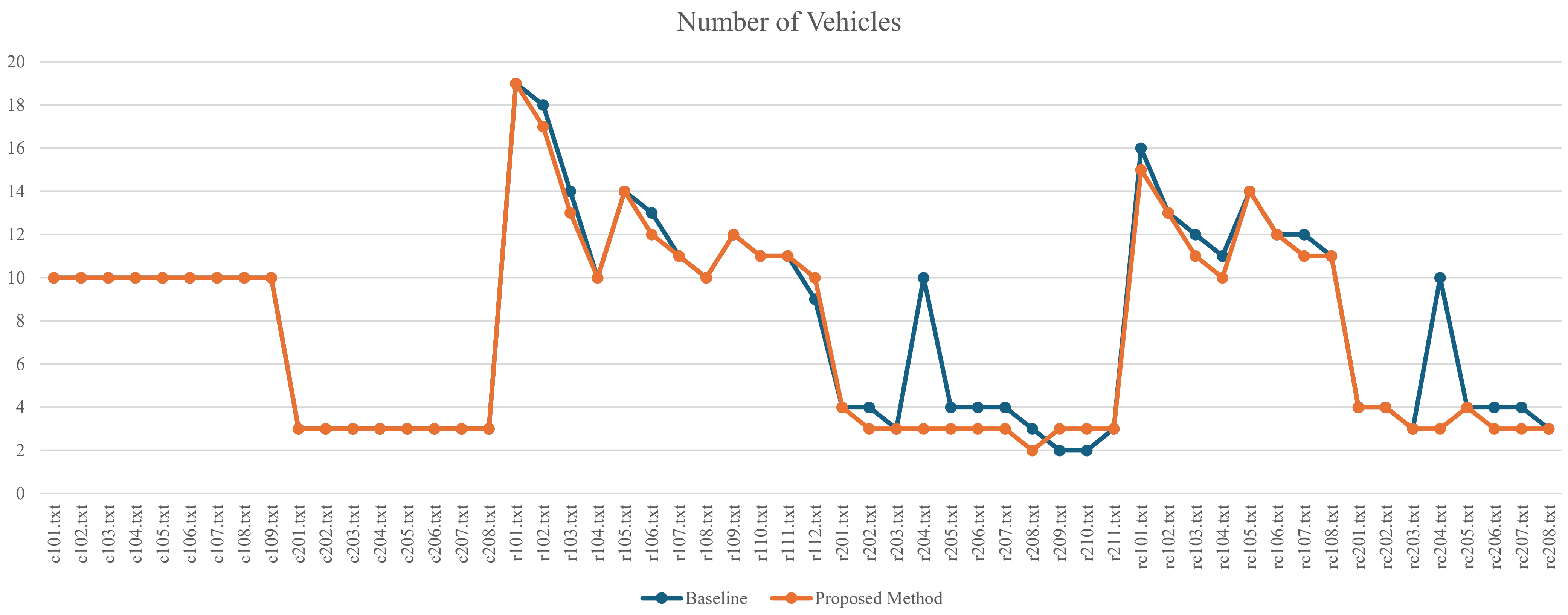
- Hybrid Genetic Algorithm-Solomon Insertion Heuristic (HGA-SIH)[5]

- Problem Definition
  - Vehicle Routing Problem with Time Windows (VRPTW)

- Method
  - Integrates the Solomon Insertion Heuristic (SIH) with a Genetic Algorithm (GA)
  - Partially Mapped Crossover (PMX) and Inverse Mutation operators

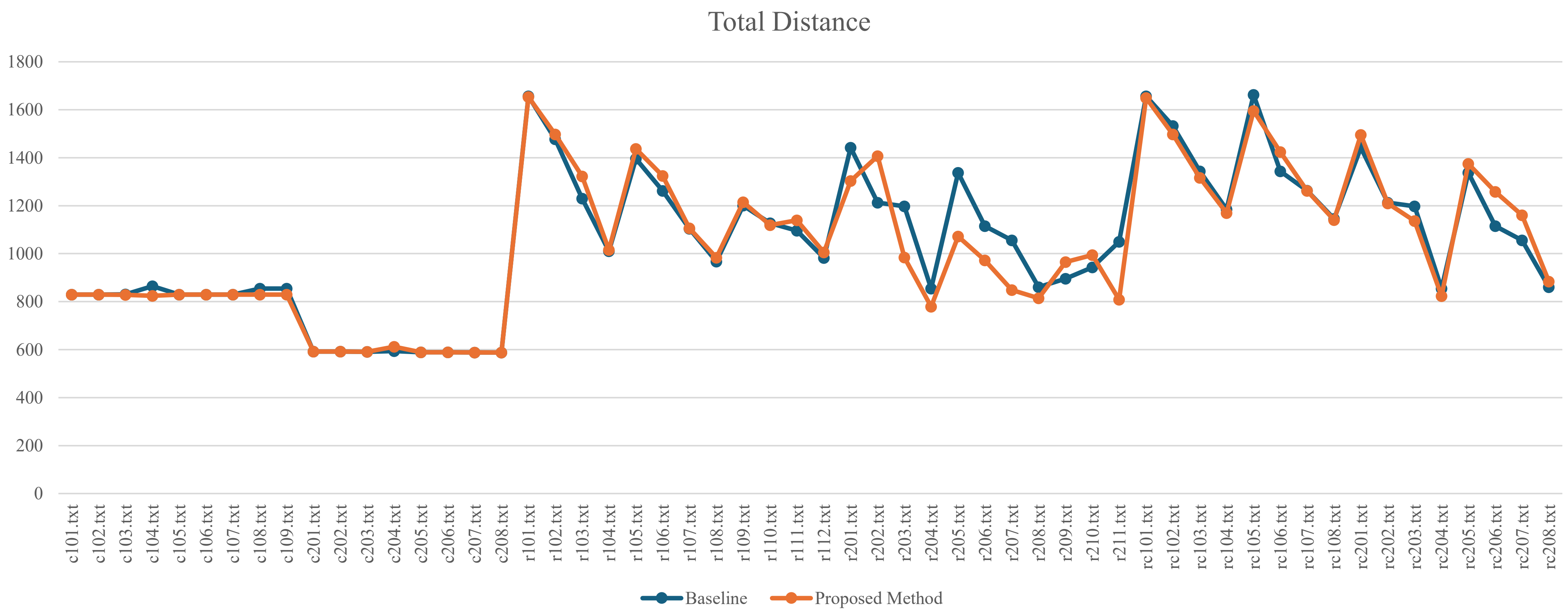
[5] A. Maroof, B. Ayvaz, and K. Naeem, "Logistics optimization using hybrid genetic algorithm (hga): A solution to the vehicle routing problem with time windows (vrptw)," *IEEE Access,* vol. 12, pp. 36974–36989, 2024

Comparison



Number of Vehicles
20
18
16
14
12
10
8
6
4
2
0
c101.txt
c102.txt
c103.txt
c104.txt
c105.txt
c106.txt
c107.txt
c108.txt
c109.txt
c201.txt
c202.txt
c203.txt
c204.txt
c205.txt
c206.txt
c207.txt
c208.txt
r101.txt
r102.txt
r103.txt
r104.txt
r105.txt
r106.txt
r107.txt
r108.txt
r109.txt
r110.txt
r111.txt
r112.txt
r201.txt
r202.txt
r203.txt
r204.txt
r205.txt
r206.txt
r207.txt
r208.txt
r209.txt
r210.txt
r211.txt
rc101.txt
rc102.txt
rc103.txt
rc104.txt
rc105.txt
rc106.txt
rc107.txt
rc108.txt
rc201.txt
rc202.txt
rc203.txt
rc204.txt
rc205.txt
rc206.txt
rc207.txt
rc208.txt
Baseline
Proposed Method

Comparison



Total Distance
1800
1600
1400
1200
1000
800
600
400
200
0
c101.txt c102.txt c103.txt c104.txt c105.txt c106.txt c107.txt c108.txt c109.txt c201.txt c202.txt c203.txt c204.txt c205.txt c206.txt c207.txt c208.txt r101.txt r102.txt r103.txt r104.txt r105.txt r106.txt r107.txt r108.txt r109.txt r110.txt r111.txt r112.txt r201.txt r202.txt r203.txt r204.txt r205.txt r206.txt r207.txt r208.txt r209.txt r210.txt r211.txt rc101.txt rc102.txt rc103.txt rc104.txt rc105.txt rc106.txt rc107.txt rc108.txt rc201.txt rc202.txt rc203.txt rc204.txt rc205.txt rc206.txt rc207.txt rc208.txt
Baseline
Proposed Method

Comparison

- Wilcoxon Signed-Rank Test
  - Significant at $\alpha=0.05$

Metric	**p-value**
Number of Vehicles	0.002741
Total Distance	0.700151

Outline

- Motivation
- The proposed approach
- Experiment
- Discussion
- Conclusion & Future

Discussion

- Problem Structure Difference
  - Fixed-route structure derived from logistics operations
  - Different from VRP with global route re-optimization
  - Benchmark evaluation focuses on support route planning

Outline

- Motivation
- The proposed approach
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- Conclusion & Future

Conclusion

- Proposed a Multi-Stage Operationally Constrained VRP model for real-world logistics
- Developed a hybrid solution framework using CCGA, HACO, and MACS
- Reduced support-route usage and improved delivery efficiency on real-world data

Future

- Problem-level Extensions
  - Multi-type Heterogeneous Fleet Vehicle Routing Problem
  - Green Vehicle Routing Problem

- Advanced Methods
  - Integrate advanced artificial intelligence methodologies
  - Apply deep learning and reinforcement learning to improve scalability and decision quality

Q & A

Thank you for listening !